In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv(r"C:\Users\HARI GOWTHAM\Downloads\archive (3)\india_job_market_2024_2026.csv")
```

In [2]:
```
df.head()
```

Out[2]:

	Job_ID	Job_Title	Company	Company_Type	Industry	City	Location_Tier	Experience_Level	Job_Type	Work_Mode	Sal
0	IND2025000	Android Developer	Tech Mahindra	MNC	Information Technology	Remote	Remote	Senior (6-10 yrs)	Full-Time	Remote	
1	IND2025001	QA Engineer	Dream11	Indian Unicorn	Information Technology	Lucknow	Tier 2	Senior (6-10 yrs)	Full-Time	Hybrid	
2	IND2025002	Business Analyst	HAL	PSU/Govt	EdTech	Remote	Remote	Senior (6-10 yrs)	Full-Time	Remote	
3	IND2025003	Cybersecurity Analyst	Groww	Startup	Information Technology	Mumbai	Tier 1	Mid (3-6 yrs)	Full-Time	Hybrid	
4	IND2025004	Python Developer	Oracle	MNC	EdTech	Remote	Remote	Junior (1-3 yrs)	Full-Time	Remote	

In [3]:
```
#columns names
df.columns
```

Out[3]:
```
Index(['Job_ID', 'Job_Title', 'Company', 'Company_Type', 'Industry', 'City',
       'Location_Tier', 'Experience_Level', 'Job_Type', 'Work_Mode',
       'Salary_LPA', 'Skills_Required', 'Education_Required', 'Openings',
       'Applicants', 'Company_Rating', 'Date_Posted'],
      dtype='object')
```

In [4]:
```
#index range
df.index
```

Out[4]:
```
RangeIndex(start=0, stop=5000, step=1)
```

In [5]:
```
#datatypes
df.dtypes
```

Out[5]:
```
Job_ID                 object
Job_Title              object
Company                object
Company_Type           object
Industry               object
City                   object
Location_Tier          object
Experience_Level       object
Job_Type               object
Work_Mode              object
Salary_LPA            float64
Skills_Required        object
Education_Required     object
Openings                int64
Applicants              int64
Company_Rating        float64
Date_Posted            object
dtype: object
```

In [6]:
```
#dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 17 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   Job_ID              5000 non-null   object 
 1   Job_Title           5000 non-null   object 
 2   Company             5000 non-null   object 
 3   Company_Type        5000 non-null   object 
 4   Industry            5000 non-null   object 
 5   City                5000 non-null   object 
 6   Location_Tier       5000 non-null   object 
 7   Experience_Level    5000 non-null   object 
 8   Job_Type            5000 non-null   object 
 9   Work_Mode           5000 non-null   object 
 10  Salary_LPA          5000 non-null   float64
 11  Skills_Required     5000 non-null   object 
 12  Education_Required  5000 non-null   object 
 13  Openings            5000 non-null   int64  
 14  Applicants          5000 non-null   int64  
 15  Company_Rating      5000 non-null   float64
 16  Date_Posted         5000 non-null   object 
dtypes: float64(2), int64(2), object(13)
memory usage: 664.2+ KB
```

In [7]:

```
#simple maths opearions
df.describe()
```

Out[7]:

	Salary_LPA	Openings	Applicants	Company_Rating
count	5000.000000	5000.000000	5000.000000	5000.000000
mean	19.829440	3.642600	302.072000	3.698420
std	18.136741	4.046942	363.989613	0.424994
min	0.800000	1.000000	14.000000	2.500000
25%	6.800000	1.000000	99.000000	3.400000
50%	13.600000	2.000000	185.000000	3.800000
75%	25.600000	3.000000	321.000000	4.100000
max	115.400000	20.000000	2387.000000	4.300000

In [8]:

```
#checking blank values
df.isna().sum()
```

Out[8]:

```
Job_ID                0
Job_Title             0
Company               0
Company_Type          0
Industry              0
City                  0
Location_Tier         0
Experience_Level      0
Job_Type              0
Work_Mode             0
Salary_LPA            0
Skills_Required       0
Education_Required    0
Openings              0
Applicants            0
Company_Rating        0
Date_Posted           0
dtype: int64
```

In [9]:
```
#checking duplicates
df.duplicated().sum()
```

Out[9]:
```
0
```

In [10]:
```
#total complanies
df['Company'].value_counts().count()
```

Out[10]:
```
53
```

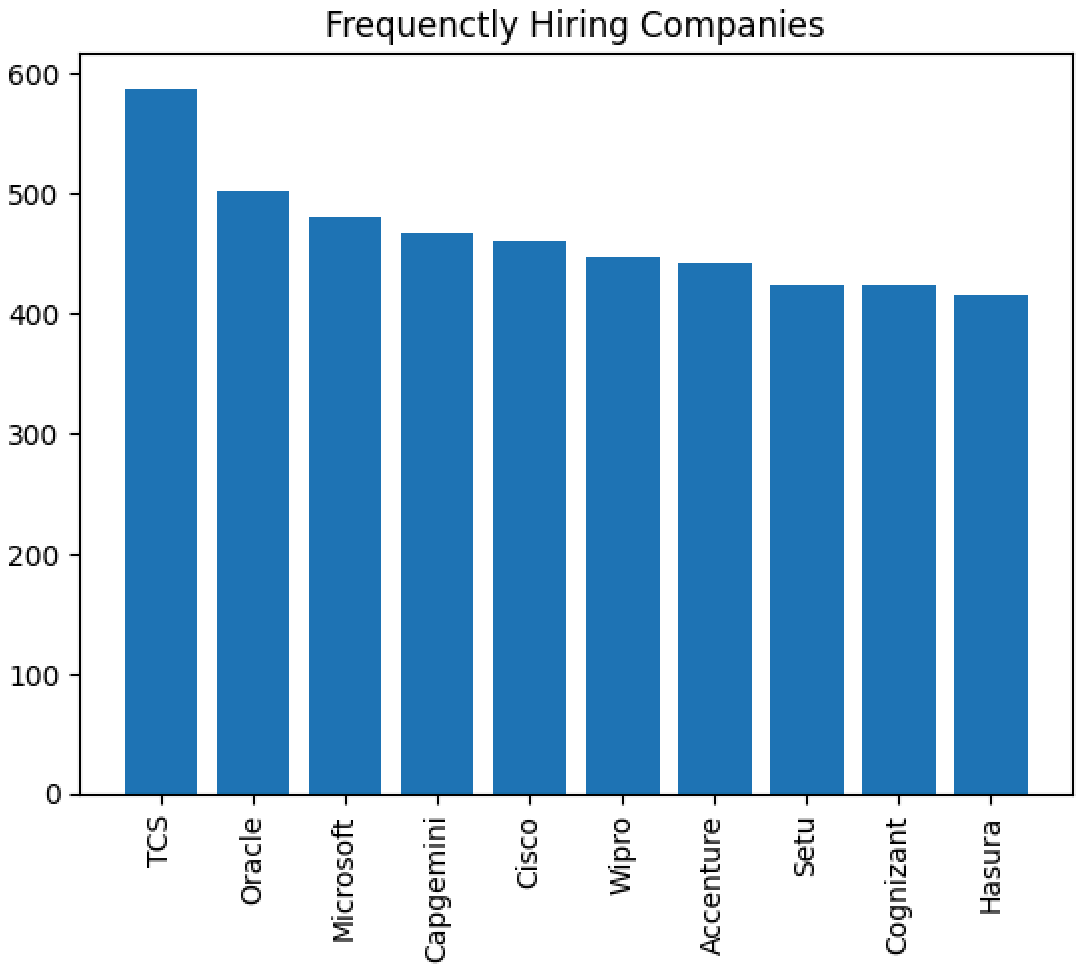
In [11]:
```
#work modes
df['Work_Mode'].value_counts()
```

Out[11]:
```
Work_Mode
On-Site    2005
Hybrid     1728
Remote     1267
Name: count, dtype: int64
```

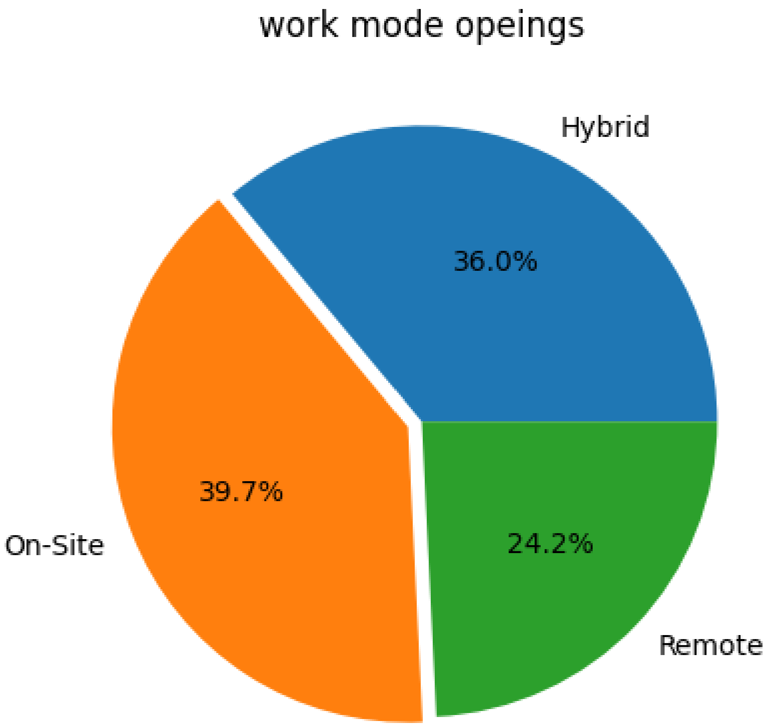
In [12]:
```
#experince category
df['Experience_Level'].value_counts()
```

Out[12]:
```
Experience_Level
Mid (3-6 yrs)        1378
Junior (1-3 yrs)     1256
Fresher (0-1 yr)     1022
Senior (6-10 yrs)     831
Lead (10+ yrs)        513
Name: count, dtype: int64
```

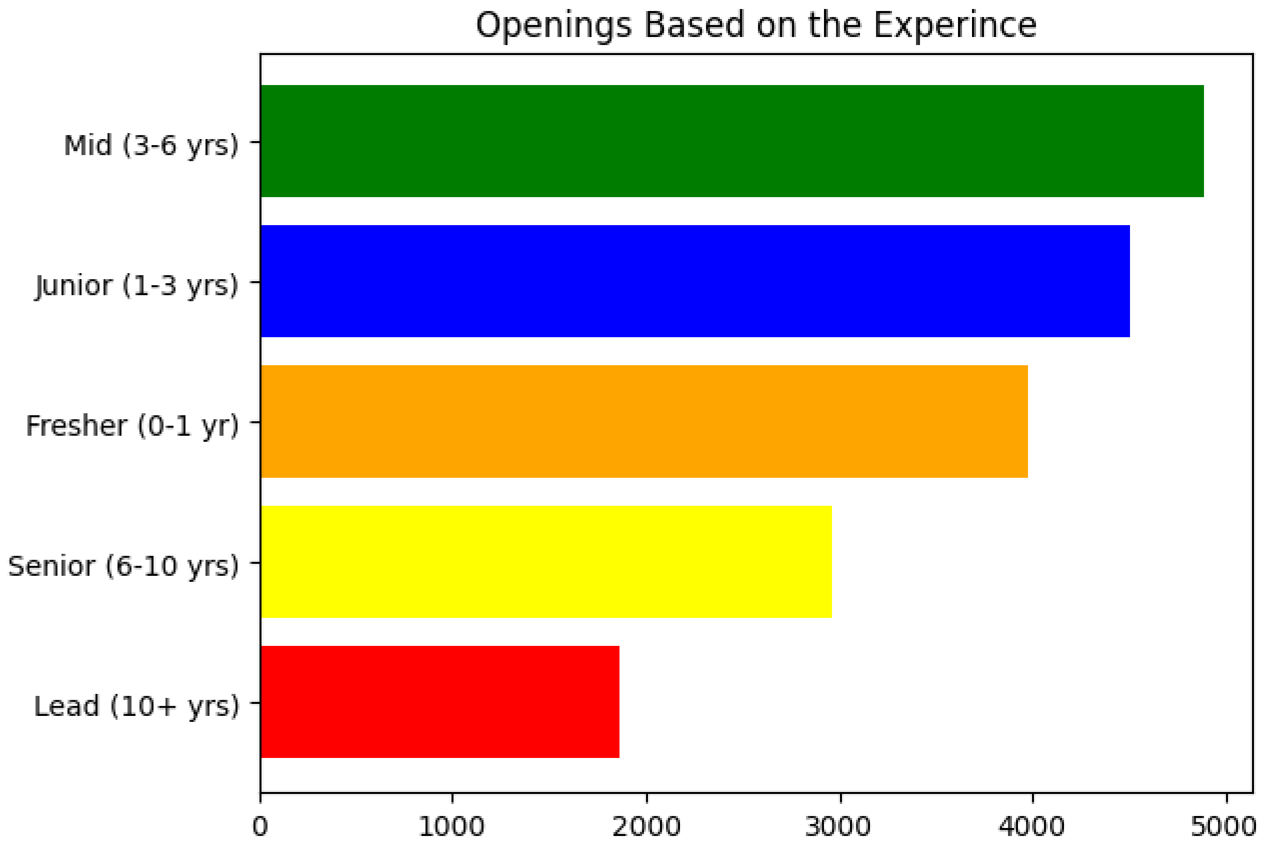
In [13]:
```
#Top 10 Companies with opeings
com_open=df.groupby('Company')['Openings'].sum().sort_values(ascending=False).head(10)
plt.title('Frequenctly Hiring Companies')
plt.bar(com_open.index,com_open.values)
plt.xticks(rotation=90)
plt.show()
```



In [14]:
```
#work mode openings
work_open=df.groupby('Work_Mode')['Openings'].sum()
plt.title('work mode opeings')
plt.pie(work_open.values,labels=work_open.index,autopct="%1.1f%%",explode=[0,0.05,0])
plt.show()
```



In [15]:
```
#experince wise openings
plt.title('Openings Based on the Experince')
exp_open=df.groupby('Experience_Level')['Openings'].sum().sort_values()
plt.barh(exp_open.index,exp_open.values,color=['red','yellow','orange','b','g'])
plt.show()
```

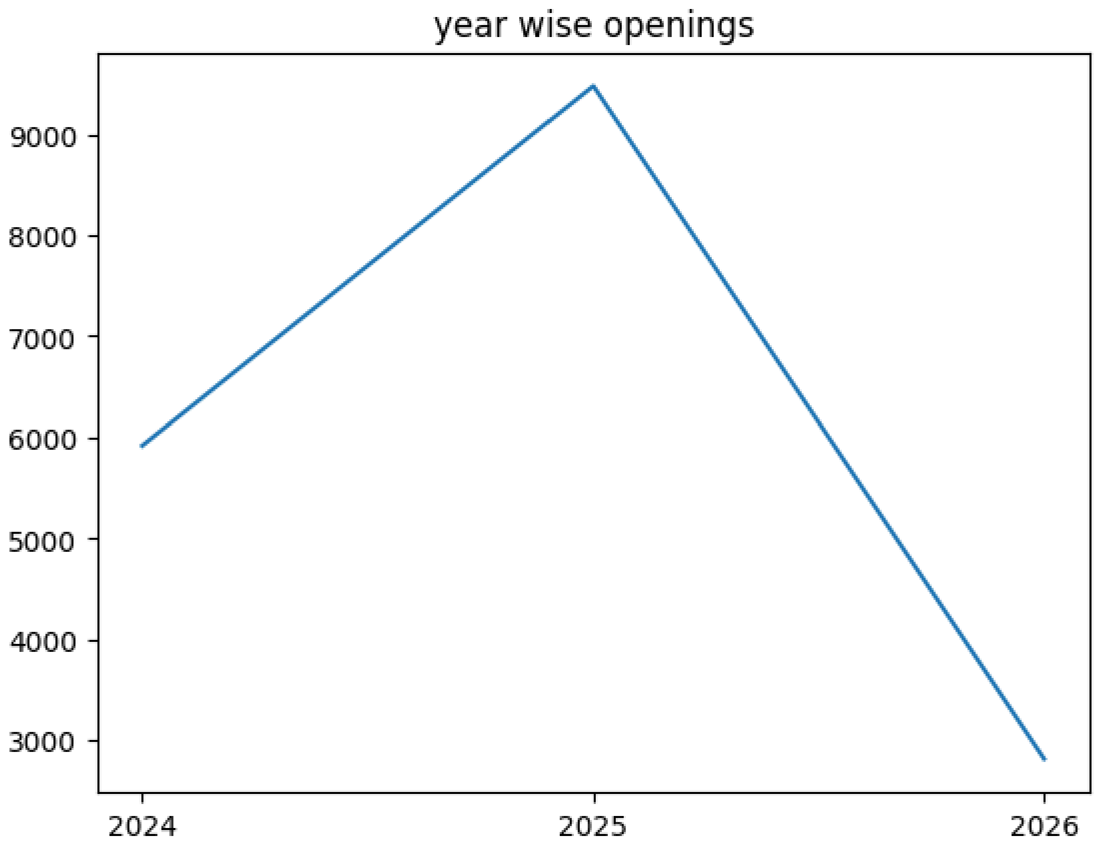


In [16]:

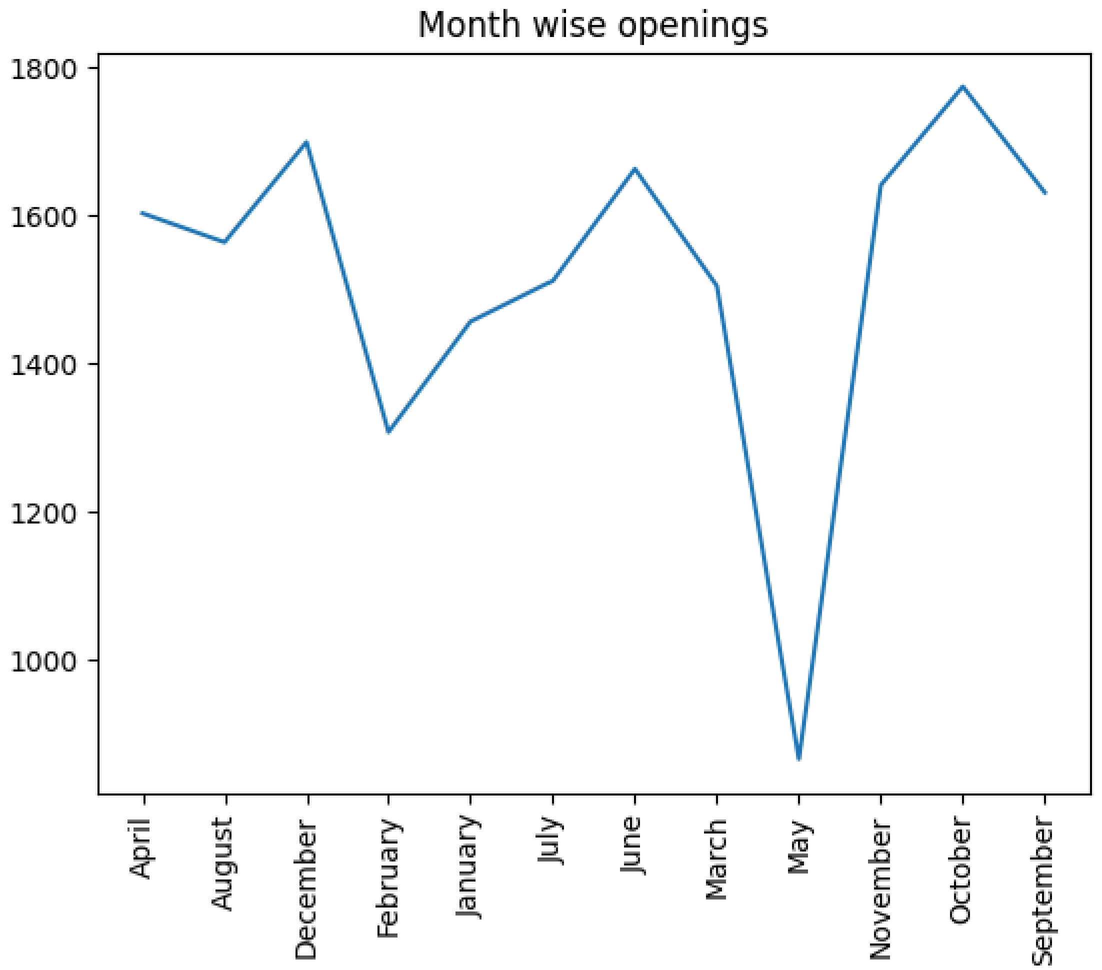
```
#comapanies for freshers opeings
freshers = df[df['Experience_Level'] == 'Fresher (0-1 yr)']
fresher_openings = freshers.groupby('Company')['Openings'].sum().sort_values(ascending=False).head(10)
c = ['#008000','#32CD32','#7CFC00','#ADFF2F','#FFFF00','#FFD700','#FFA500','#FF8C00','#FF4500','#FF0000']
plt.title('Most Freshers Hiring Comapanies')
plt.barh(fresher_openings.index,fresher_openings.values,color=c)
plt.show()
```

In [54]:
```
#convert string date format into date format
df['Date_Posted']=pd.to_datetime(df['Date_Posted'],dayfirst=True)
```

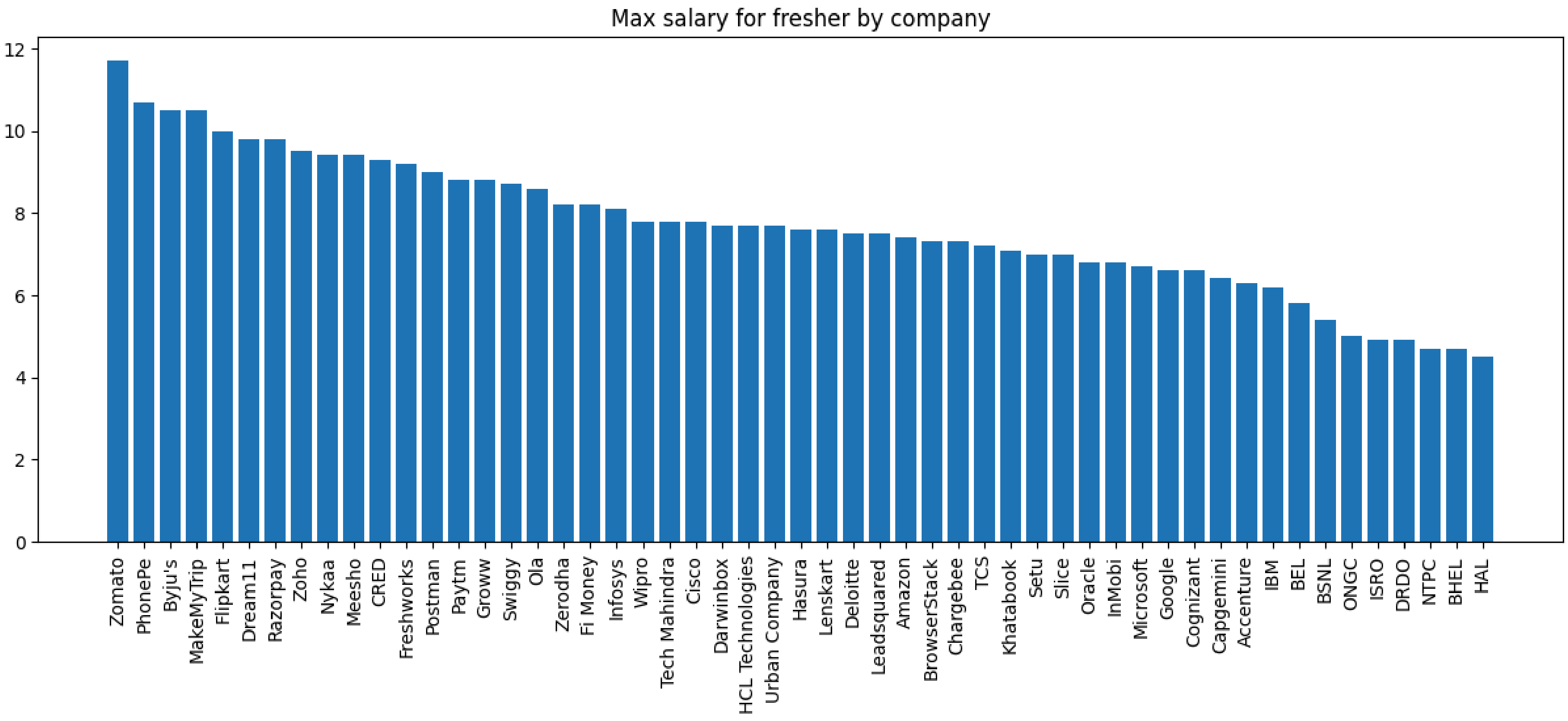
In [18]:
```
#year wise openings
year_open=df.groupby(df['Date_Posted'].dt.year)['Openings'].sum()
plt.title('year wise openings')
plt.plot(year_open.index,year_open.values)
plt.xticks([2024,2025,2026])
plt.show()
```



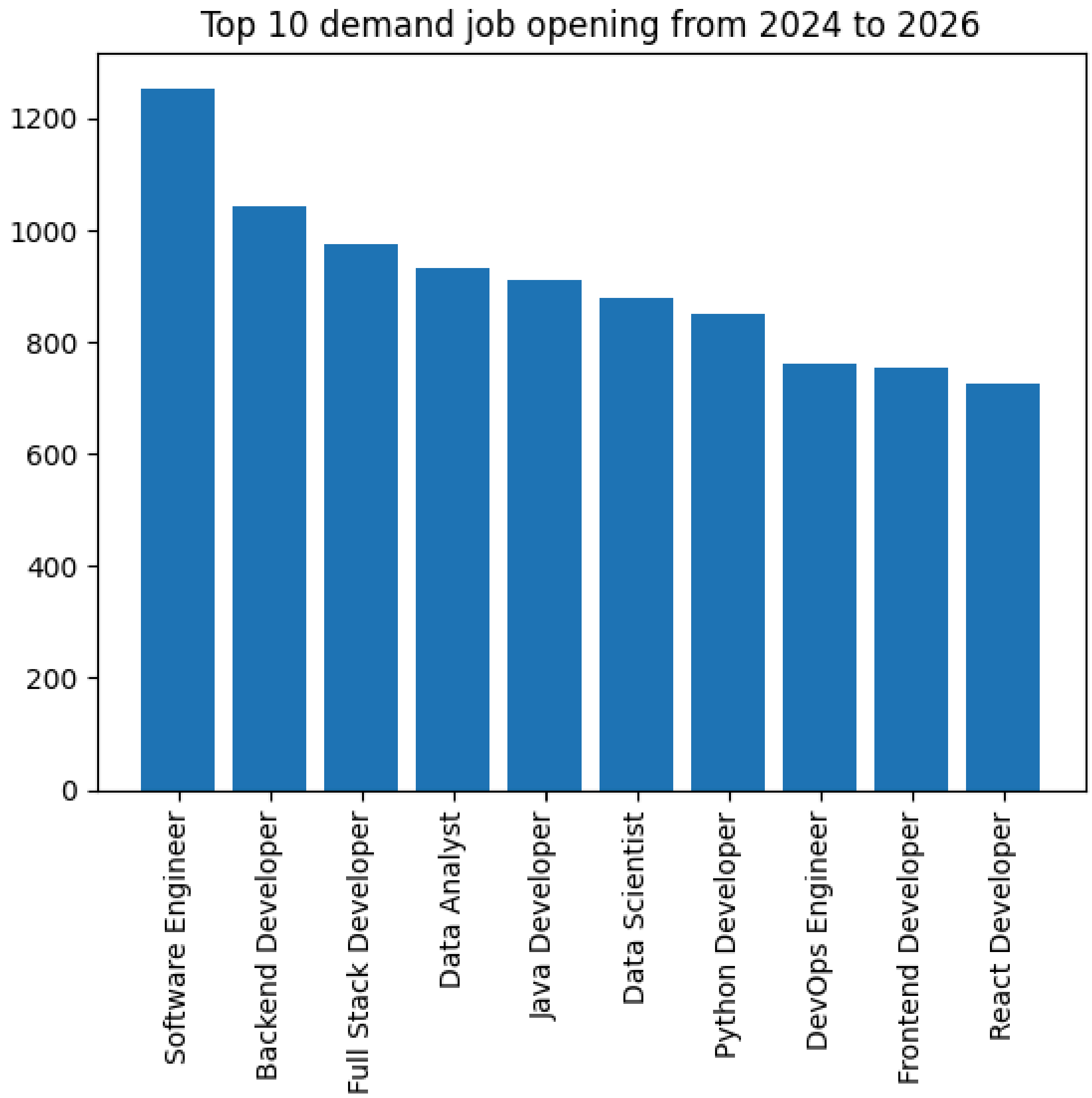
In [19]:
```
#Month wise openings
mon_open=df.groupby(df['Date_Posted'].dt.month_name())['Openings'].sum()
plt.plot(mon_open.index,mon_open.values)
plt.title('Month wise openings')
plt.xticks(rotation=90)
plt.show()
```



In [20]:
```
#Max salary for fresher by company
freshers_max_salary=freshers.groupby('Company')['Salary_LPA'].max().sort_values(ascending=False)
plt.figure(figsize=(15,5))
plt.title('Max salary for fresher by company')
plt.bar(freshers_max_salary.index,freshers_max_salary.values)
plt.xticks(rotation=90)
plt.show()
```

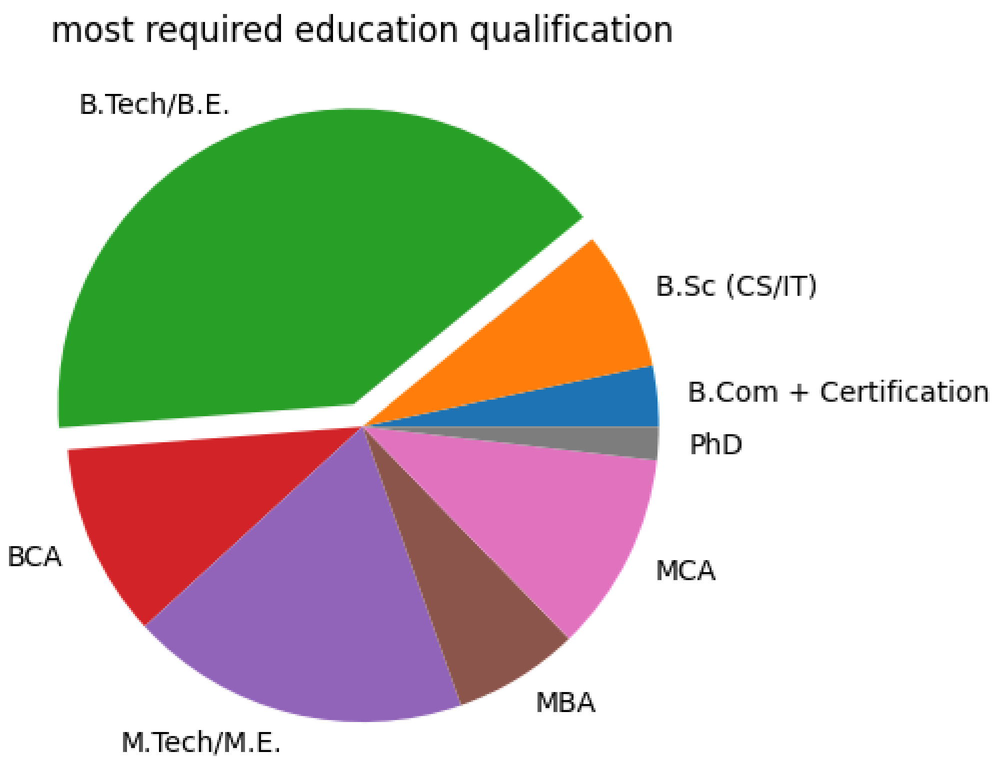


In [21]:
```
#Top 10 demand job opening from 2024 to 2026
job_open=df.groupby('Job_Title')['Openings'].sum().sort_values(ascending=False).head(10)
plt.bar(job_open.index,job_open.values)
plt.title('Top 10 demand job opening from 2024 to 2026')
plt.xticks(rotation=90)
plt.show()
```

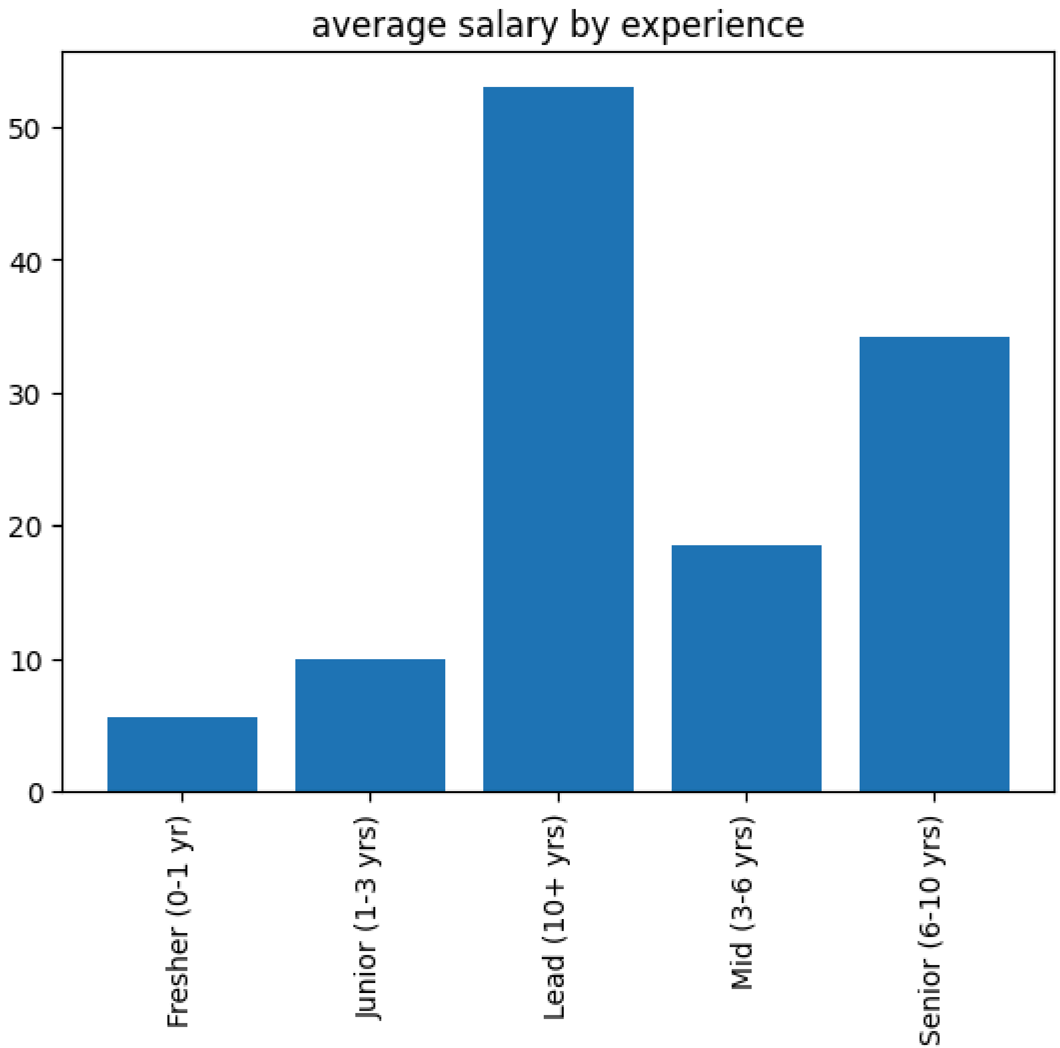


In [22]:

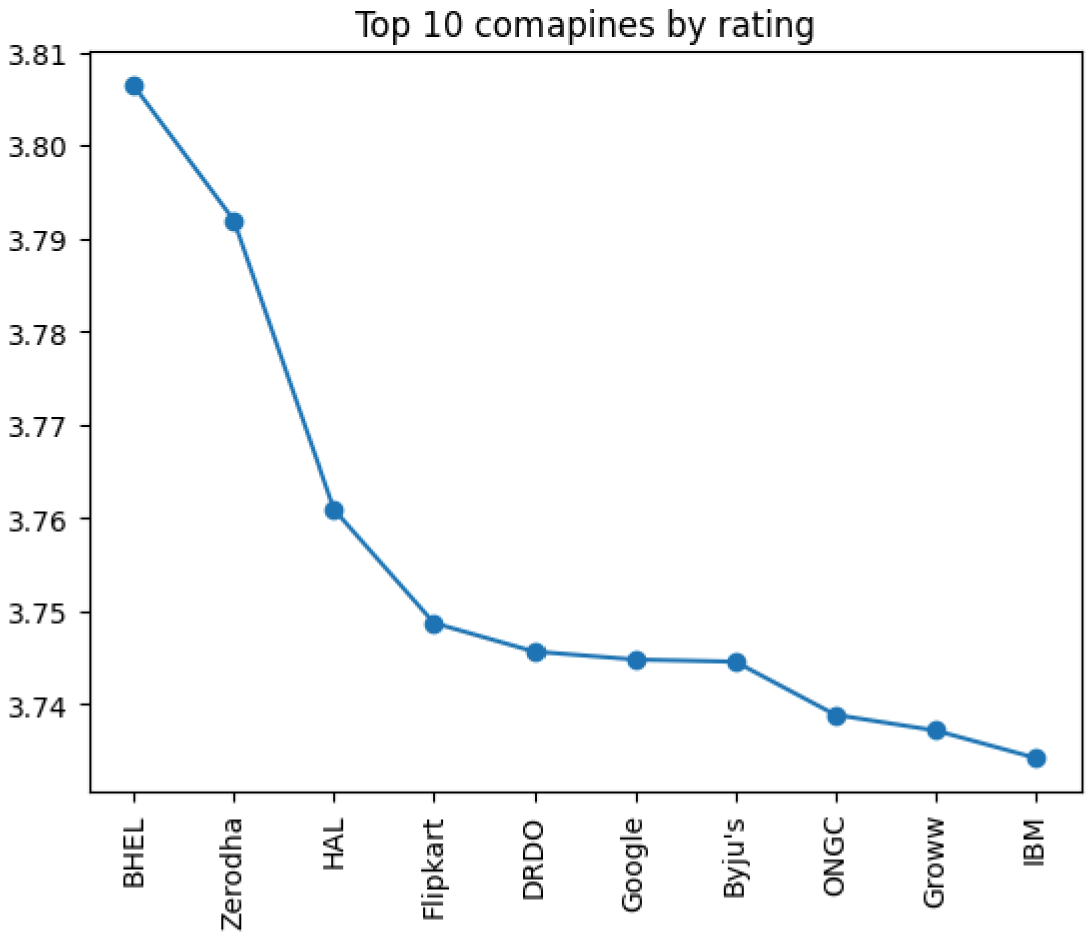
```
#Required education qualification
edu=df.groupby('Education_Required')['Openings'].sum()
plt.title('most required education qualification')
plt.pie(edu.values,labels=edu.index,explode=[0,0,0.08,0,0,0,0,0])
plt.show()
```



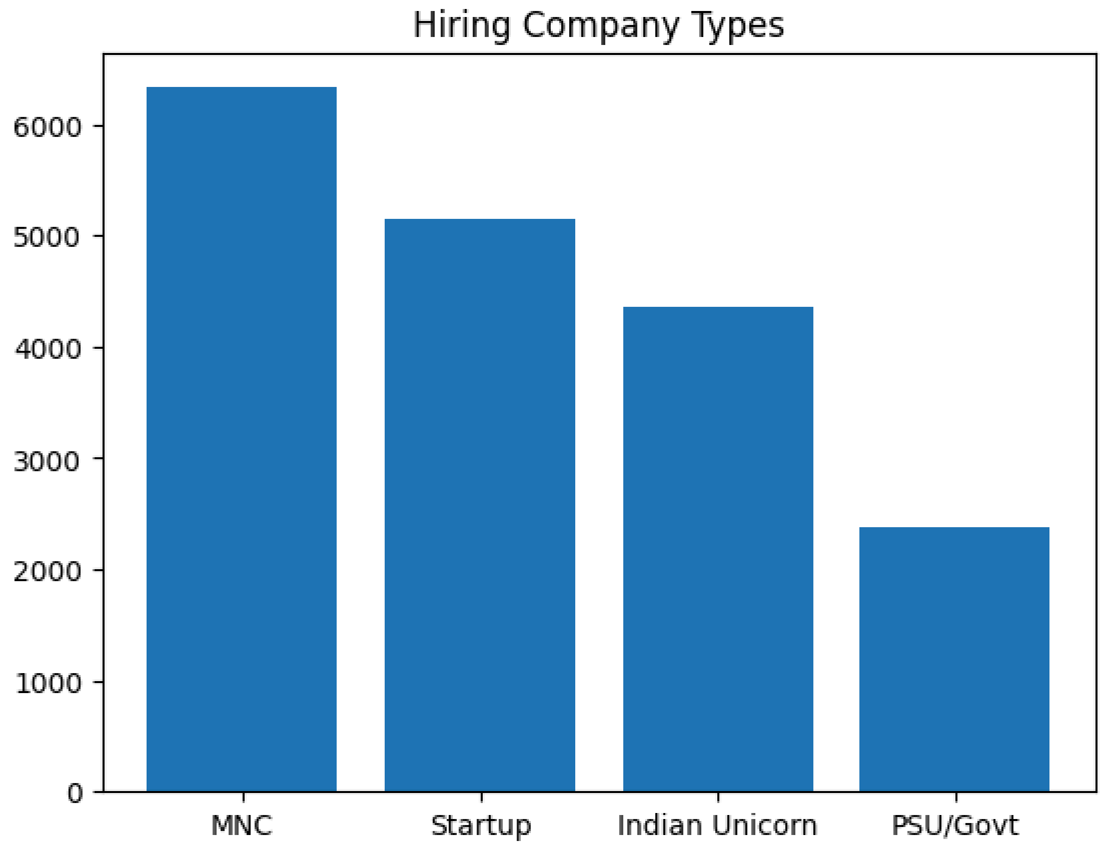
In [23]:
```
#average salary by experience
avg_salary=df.groupby('Experience_Level')['Salary_LPA'].mean()
plt.title('average salary by experience')
plt.bar(avg_salary.index,avg_salary.values)
plt.xticks(rotation=90)
plt.show()
```



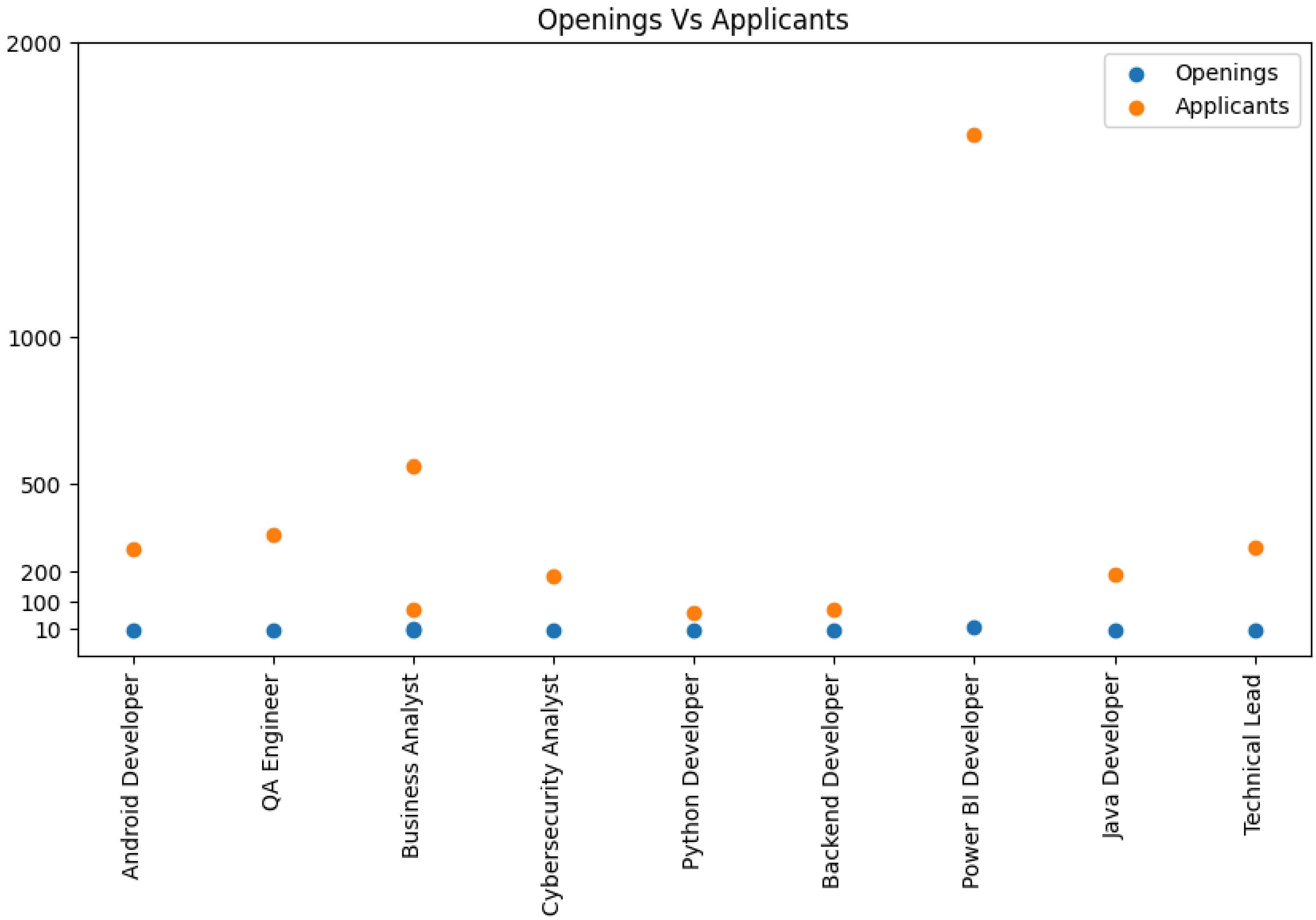
In [24]:
```
#Top 10 comapines by rating
com_rating=df.groupby('Company')['Company_Rating'].mean().sort_values(ascending=False).head(10)
plt.plot(com_rating.index,com_rating.values,marker='o')
plt.title('Top 10 comapines by rating')
plt.xticks(rotation=90)
plt.show()
```



In [26]:
```
#hring comapny types
com_type_openings=df.groupby('Company_Type')['Openings'].sum().sort_values(ascending=False)
plt.bar(com_type_openings.index,com_type_openings)
plt.title('Hiring Company Types')
plt.show()
```



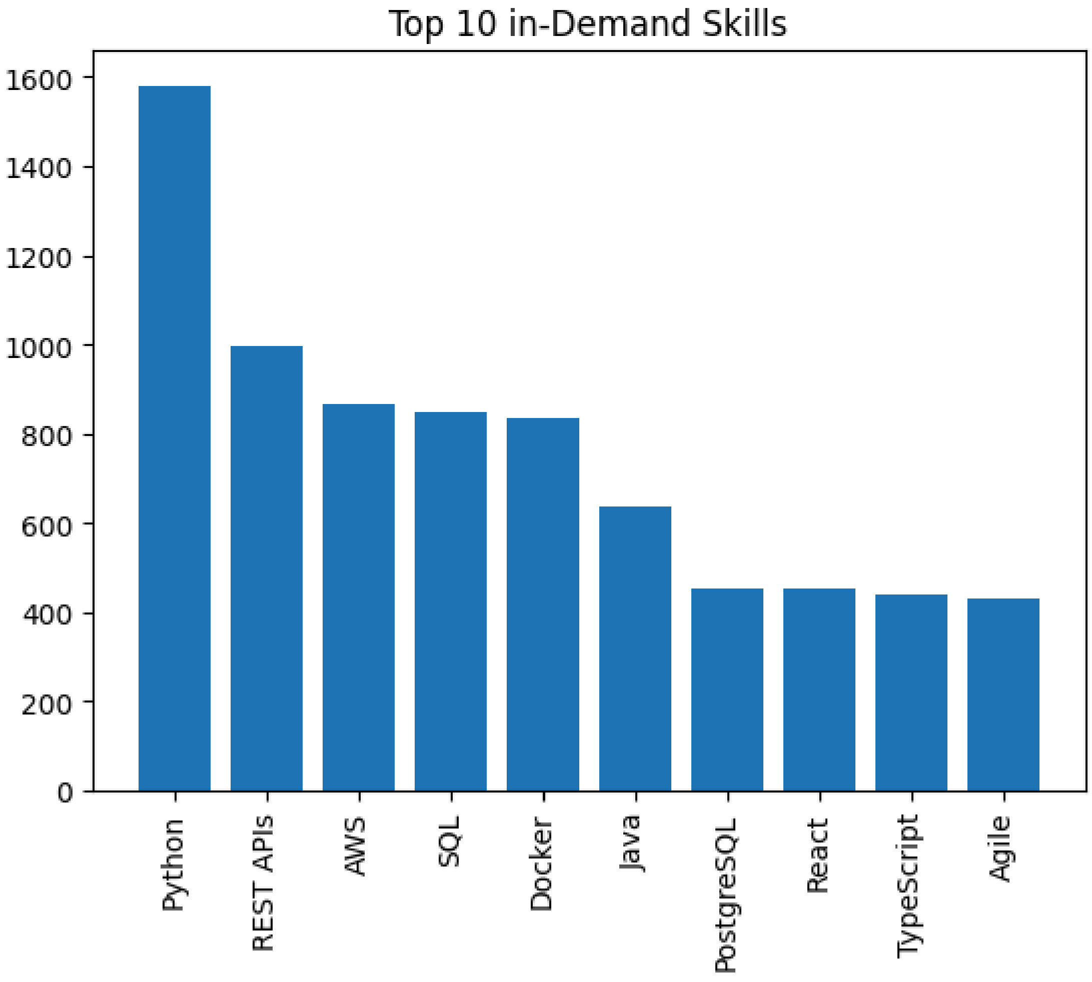
In [56]:
```
plt.figure(figsize=(10,5))
plt.scatter(df['Job_Title'].head(10),df['Openings'].head(10),label='Openings')
plt.scatter(df['Job_Title'].head(10),df['Applicants'].head(10),label='Applicants')
plt.xticks(rotation=90)
plt.yticks([10,100,200,500,1000,2000])
plt.legend()
plt.title('Openings Vs Applicants')
plt.show()
```



In [69]:

```
#Most Popular Skill
skills_count = (
    df['Skills_Required']
    .str.split(',')
    .explode()
```

```
    .str.strip()
    .value_counts()
).sort_values(ascending=False).head(10)
plt.bar(skills_count.index,skills_count.values)
plt.xticks(rotation=90)
plt.title('Top 10 in-Demand Skills')
plt.show()
```



In [ ]: